Jumpei F. Yamagishi

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Experience

April 2024 – Present: Special Postdoctoral Researcher (SPDR), RIKEN, Japan.

Education

March 2024: Ph.D. in Arts and Sciences, The University of Tokyo, Tokyo, Japan.
Advisors: Kunihiko Kaneko, Shuji Ishihara

March 2021: Master in Arts and Sciences, The University of Tokyo, Tokyo, Japan.

March 2019: Bachelor in Arts and Sciences, The University of Tokyo, Tokyo, Japan.

Papers (Peer-Reviewed)

1. **Jumpei F. Yamagishi** and Tetsuhiro S. Hatakeyama. Global Constraint Principle for Microbial Growth Laws. Proceedings of the National Academy of Sciences of the United States of America (PNAS), 122 (40), e2515031122, 2025.
2. **Jumpei F. Yamagishi**, Kunihiko Kaneko. Universal Transitions between Growth and Dormancy via Intermediate Complex Formation. Physical Review Letters, 132, 118401, 2024.
3. **Jumpei F. Yamagishi** and Tetsuhiro S. Hatakeyama. Linear Response Theory of Evolved Metabolic Systems. Physical Review Letters 131, 028401, 2023.
4. Masahiro Hoshino, Ryuna Nagayama, Kohei Yoshimura, **Jumpei F. Yamagishi**, and Sosuke Ito. A geometric speed limit for acceleration by natural selection in evolutionary processes. Physical Review Research 5, 023127, 2023.
5. **Jumpei F. Yamagishi** and Tetsuhiro S. Hatakeyama. Microeconomics of metabolism: The Warburg effect as Giffen behaviour. Bulletin of Mathematical Biology, 83: 120, 2021.
6. **Jumpei F. Yamagishi**, Nen Saito, and Kunihiko Kaneko. Adaptation of metabolite leakiness leads to symbiotic chemical exchange and to a resilient microbial ecosystem. PLoS Computational Biology 17(6): e1009143, 2021.
7. **Jumpei F. Yamagishi**, Nen Saito, and Kunihiko Kaneko. Advantage of leakage of essential metabolites for cells. Physical Review Letters 124, 048101, 2020.
8. **Jumpei F. Yamagishi** and Kunihiko Kaneko. Chaos with a high-dimensional torus. Physical Review Research 2, 023044, 2020.

9. **Jumpei F. Yamagishi**, Nen Saito, and Kunihiko Kaneko. Symbiotic Cell Differentiation and Cooperative Growth in Multicellular Aggregates. PLoS Computational Biology 12(10), e1005042, 2016.

Review Article (Invited)

10. **Jumpei F. Yamagishi** and Tetsuhiro S. Hatakeyama. Microeconomics of Cellular Metabolism. Economic Principles in Cell Biology, 2025. (To appear)

    Review article on the "microeconomics of metabolism," based on my previous work [1,3,5]. Part of the international open-access textbook project, "Economic Principles in Cell Biology" (https://doi.org/10.5281/zenodo.12592398).

Honors and Awards

March 2024: Young Scientist Award of the Physical Society of Japan.

February 2024: JSPS Ikushi Prize, Japan Society for the Promotion of Science.

> The Ikushi Prize aims to recognize outstanding doctoral students who show promise of contributing to Japan's future scientific advancement, and was established by JSPS in 2010 with an endowment from His Majesty the Emperor Emeritus Akihito.

September 2022: Early Career Presentation Award, The Biophysical Society of Japan.

March 2019: The University of Tokyo President's Award (Grand Prize).

> Only one student is selected for the Grand Prize from among all graduates (undergraduate, master's, and doctoral) at the University of Tokyo that year.

September 2022: IUPAB (International Union for Pure and Applied Biophysics) Student Award.

March 2021: Senko-Shorei Sho, Graduate School of Arts and Sciences, The University of Tokyo.

March 2021: Student Presentation Award (Division 12), Tthe Physical Society of Japan.

March 2020: Student Presentation Award (Division 11), Tthe Physical Society of Japan.

Fellowships and Funding

October 2025 - March 2028 : ACT-X [Life and Information], Japan Science and Technology Agency (JST).

April 2021 - March 2024 : Japan Society for the Promotion of Science (JSPS) Research Fellowships for Young Scientists DC1.

July 2020 - February 2025 : The Masason Foundation Fellowship.

International Conferences (Invited)

1. Jumpei F. Yamagishi. Microeconomic Principles of Cellular Metabolism and Growth. International Symposium of Life-Nonlife Transition, March 18-19, 2026, Tokyo, Japan. (Scheduled)

2. Jumpei F. Yamagishi. Microeconomics of Metabolism. 1st India-Japan Workshop on Physical Aspects of Living Systems. Tokyo, Japan, February 20, 2025.
3. Jumpei F. Yamagishi. Microeconomics of Metabolism: The Warburg effect as Giffen behavior. International Workshop on Multi-scale Biological Plasticity, January 8, 2024, Tokyo, Japan.

International Conferences (Contributed Talk)

4. Jumpei F. Yamagishi. Microeconomics of Cellular Metabolism. Microbial Communities: Energetics and Dynamics Across Space and Time, October 27-31, 2025, Chicago, USA.
5. Jumpei F. Yamagishi and Tetsuhiro S. Hatakeyama. Microeconomics of Metabolism: A Linear Response Theory of Evolved Metabolic Systems. APS March Meeting 2024, March 3-8, 2024, Minneapolis, USA.
6. Jumpei F. Yamagishi and Tetsuhiro S. Hatakeyama. Microeconomics of Metabolism: A Linear Response Theory of Evolved Metabolic Systems. ICBP 2023, August 14-18, 2023, Seoul, Korea.
7. Jumpei F. Yamagishi and Tetsuhiro S. Hatakeyama. Microeconomics of Metabolism: A Linear Response Theory of Evolved Metabolic Systems. Statistical Physics and Information-Processing in Living Systems, August 3-5, 2023, Tokyo, Japan.
8. Jumpei F. Yamagishi, Nen Saito, and Kunihiko Kaneko. Microbial Potlatch: The advantage of leakage of essential metabolites and resultant symbiosis of diverse species. IUPAB, October 4-7, 2021, Online.
9. Jumpei F. Yamagishi and Kunihiko Kaneko. Chaos with a high-dimensional torus. Dynamics Days Digital 2020, August 24-27, 2020, Online.
10. Jumpei F. Yamagishi, Nen Saito, and Kunihiko Kaneko. Cellular Potlatch: The advantage of leakage of essential metabolites and resultant symbiosis of diverse species. Workshop Evolution of interacting populations, September 15-17, 2019, Plön, Germany.
11. Jumpei F. Yamagishi and Tetsuhiro S. Hatakeyama. Microeconomics of metabolism: Overflow metabolism as Giffen behavior. Advances in Complex Systems - From Ecology to Economics, July 22-26, 2019, Como, Italy.
12. Jumpei F. Yamagishi and Tetsuhiro S. Hatakeyama. Microeconomic metabolism: an exact mapping between q-bio and microeconomics, Winter Q-Bio 2019, February 18-22, 2019, Hawaii, USA.

International Conferences (Poster)

13. Jumpei F. Yamagishi and Tetsuhiro S. Hatakeyama. Global Constraint Principle for Microbial Growth Laws. Microbial Communities: Energetics and Dynamics Across Space and Time, October 27-31, 2025, Chicago, USA.
14. Jumpei F. Yamagishi and Tetsuhiro S. Hatakeyama. Microeconomics of Metabolism. 2025 Quantitative Microbiology Symposium, March 24-25, 2025, Taipei, Taiwan.
15. Jumpei F. Yamagishi and Tetsuhiro S. Hatakeyama. Global Constraint Principle for Microbial Growth Law. 1st India-Japan Workshop on Physical Aspects of Living Systems, February 19-21, 2025, Tokyo, Japan.

16. Jumpei F. Yamagishi and Tetsuhiro S. Hatakeyama. Microeconomics of Metabolism: The Warburg effect as Giffen behavior. 45th Indian Biophysical Society Meeting, March 27-29, 2023, Bangalore, India.
17. Jumpei F. Yamagishi and Tetsuhiro S. Hatakeyama. Microeconomics of metabolism: The Warburg effect as Giffen behavior. Major ideas in quantitative microbial physiology: Past, Present and Future, June 13-15, 2022, Copenhagen, Denmark.
18. Jumpei F. Yamagishi, Nen Saito, and Kunihiko Kaneko. Promotion of cell growth by leakage of chemical components and its impact on symbiosis. Evolution of Diversity, February 25 - March 2, 2018, Les Houches, France.
19. Jumpei F. Yamagishi, Nen Saito, and Kunihiko Kaneko. Symbiotic Cell Differentiation and Cooperative Growth in Multicellular Aggregates. Winter Q-Bio 2017, February 21-24, 2017, Hawaii, USA.
20. Jumpei F. Yamagishi, Nen Saito, and Kunihiko Kaneko. Complementary Cell Differentiation and Collective Growth in Catalytic Reaction Networks. QBiC Symposium 2015, August 25-26, 2015, Osaka, Japan.

Teaching Experience

The University of Tokyo, Tokyo, Japan

Autumn 2024, 2025: Teaching Assistant, Theoretical Exercises II (Department of Physics)

Autumn 2019, 2021, 2022: Teaching Assistant, Tutorial on Mathematical Sciences I (College of Arts and Sciences)

Autumn 2019, 2020; Spring 2021, 2023: Teaching Assistant, Specialized Seminar in Universal Biology (College of Arts and Sciences)

Outreach / Broader Impact (Japanese)

Komaba Science Club, University of Tokyo (December 25, 2023): A special lecture for graduate students at the University of Tokyo entitled “Microeconomics of Metabolism: Predicting Cellular Behavior through Microeconomics."

Asahi Junior and Senior High School Newspaper (February 14, 2021): Essay in the "Inou Diary" series titled "When you find something you love, explore it with all your might”.

Haiku magazine "Manbou(翻車魚)" (Issue 3): Lead essay titled "The Science of Combinations”.

Faculty of Basic Science, University of Tokyo, Department Website: Featured in a student interview.

Saturday Course for High School Students (Nada High School, October 5, 2019): A special lecture for junior and senior high school students, titled "Where Physics, Biology, and Economics Intersect".

"Komaba Style" published by the Faculty of Basic Science, University of Tokyo (University of Tokyo Press): Column in the student's voice discussing "The Stability of Ecosystems and Delving into the Dynamics of Natto.”

University of Tokyo Newspaper (June 11, 2019): "A Closer Look at the University of Tokyo President's Award Winners - Junpei Yamagishi: Interdisciplinary Approach to the Mysteries of Cells.”